

CS & IT ENGINEERING

Digital Logic



Boolean Theorems & Logical Gates

Lecture No. 08



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Recap of Previous Lecture



Topic

Introduction of Digital Logic



+
XOR

XNOR

Practice Questions

Topics to be Covered



Topic

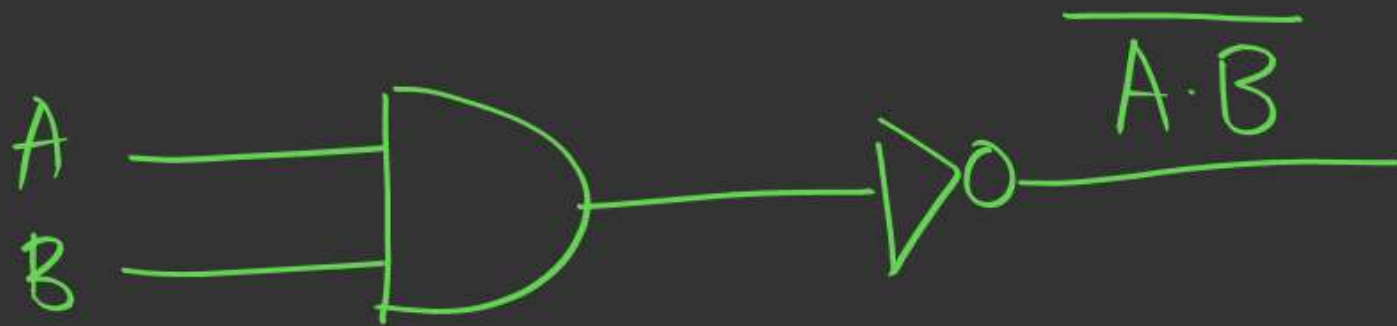
Boolean Theorem & Logic Gates



+
Universal Gates
↳ NAND
↳ NOR

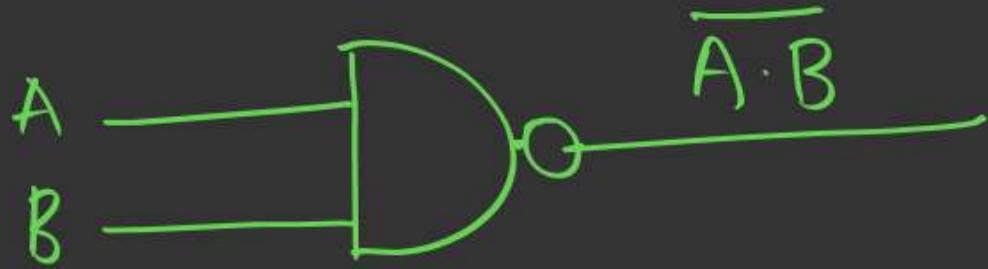
NAND gate

↓
(AND → NOT) = NAND

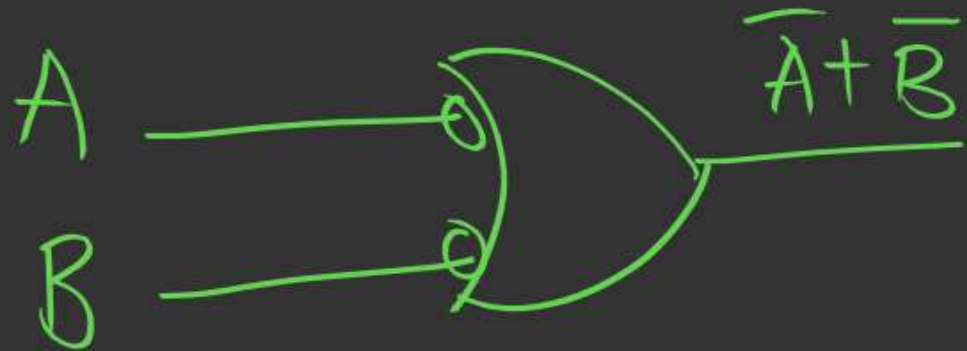


Representation

NAND :



$$y = \overline{A} + \overline{B}$$



$$\begin{aligned} y &= \overline{A \cdot B} \\ &= \overline{A} + \overline{B} \end{aligned}$$

	A	B	$\overline{A \cdot B}$
0	0	0	1
1	0	1	1
2	1	0	1
3	1	1	0

$$\underline{\underline{\text{SOP}}}$$

$$\Sigma (0, 1, 2)$$

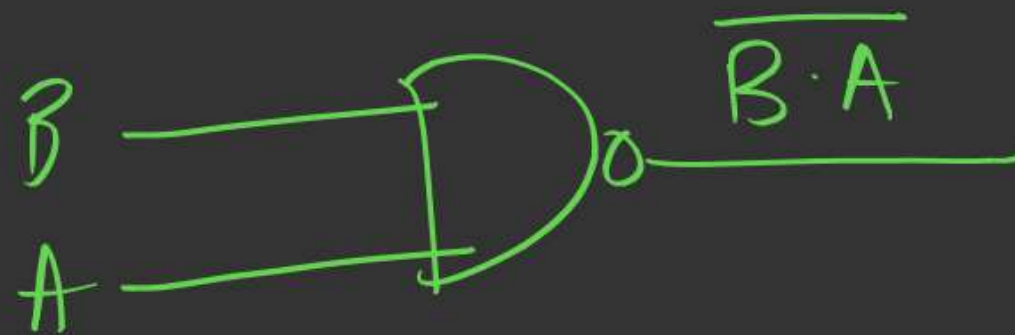
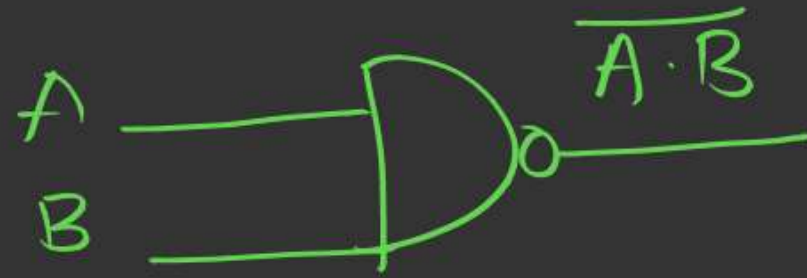
$$\text{POS} \Rightarrow \Pi (3)$$

$$\overline{A + B} = \overline{AB}$$

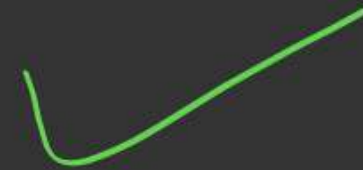
Obs :- 1) If any i/p is 0, NAND is 1.

2) When all i/p's are 1 then only NAND is 0.

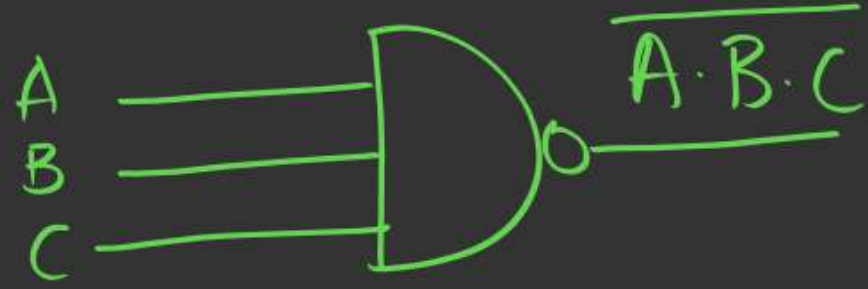
Commutative Law:-



$$\overline{A \cdot B} = \overline{B \cdot A}$$



Associative Law



Does NOT hold

Check:



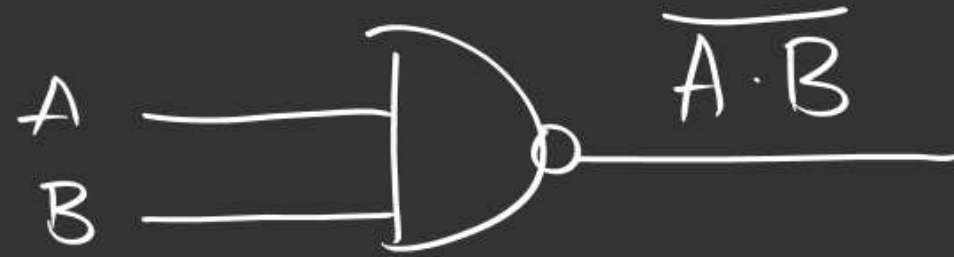
$\neq \overline{A \cdot B \cdot C}$

NAND → universal Gate



Basic : NOT, OR, AND

▷ NOT ✓

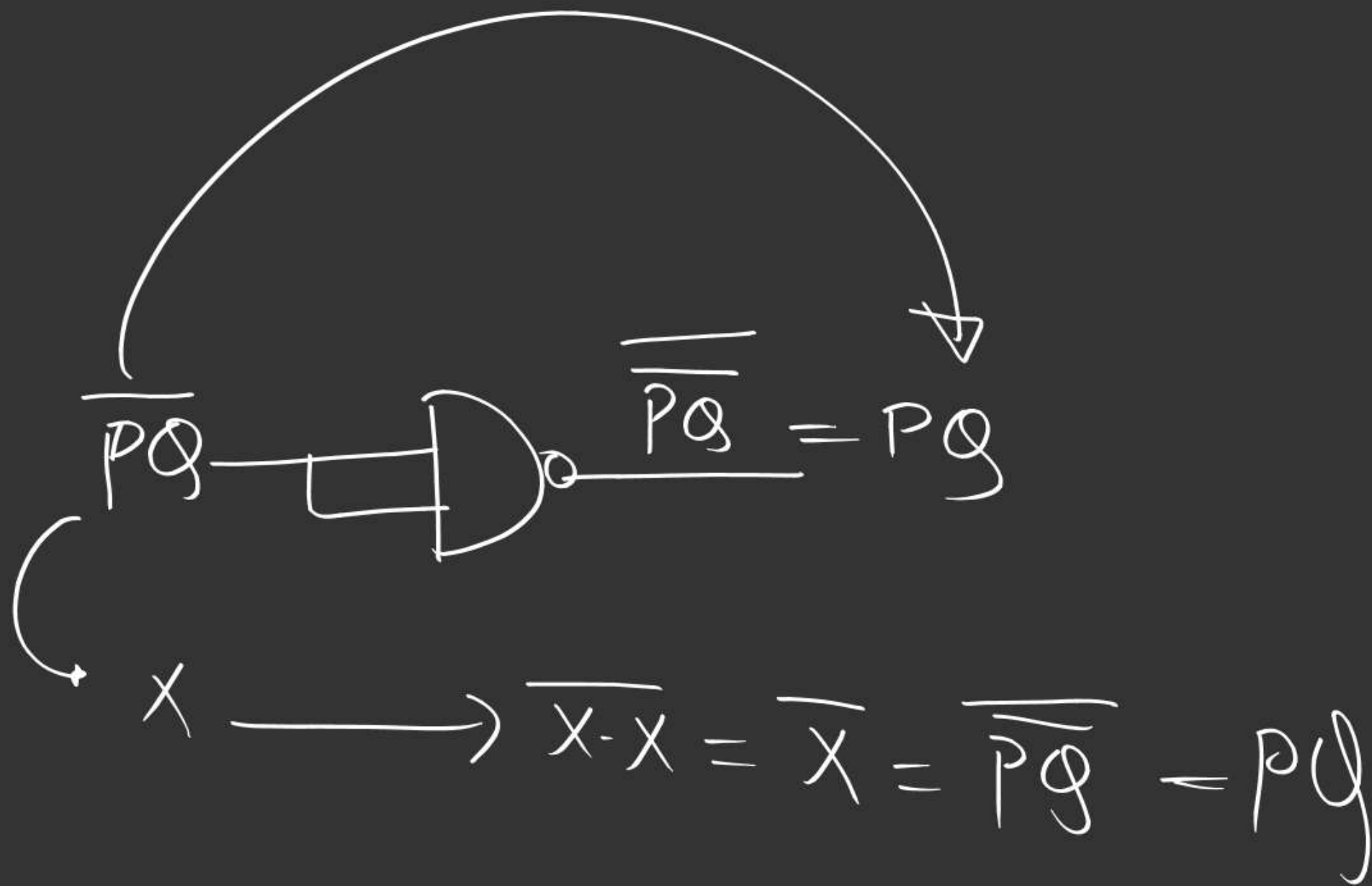


$$\overline{A \cdot A} = \overline{A}$$



$$\overline{\overline{PQ}} = PQ$$

$$\begin{array}{l} A \rightarrow \overline{\overline{A}} \\ \overline{A} \rightarrow \overline{\overline{A}} \end{array}$$



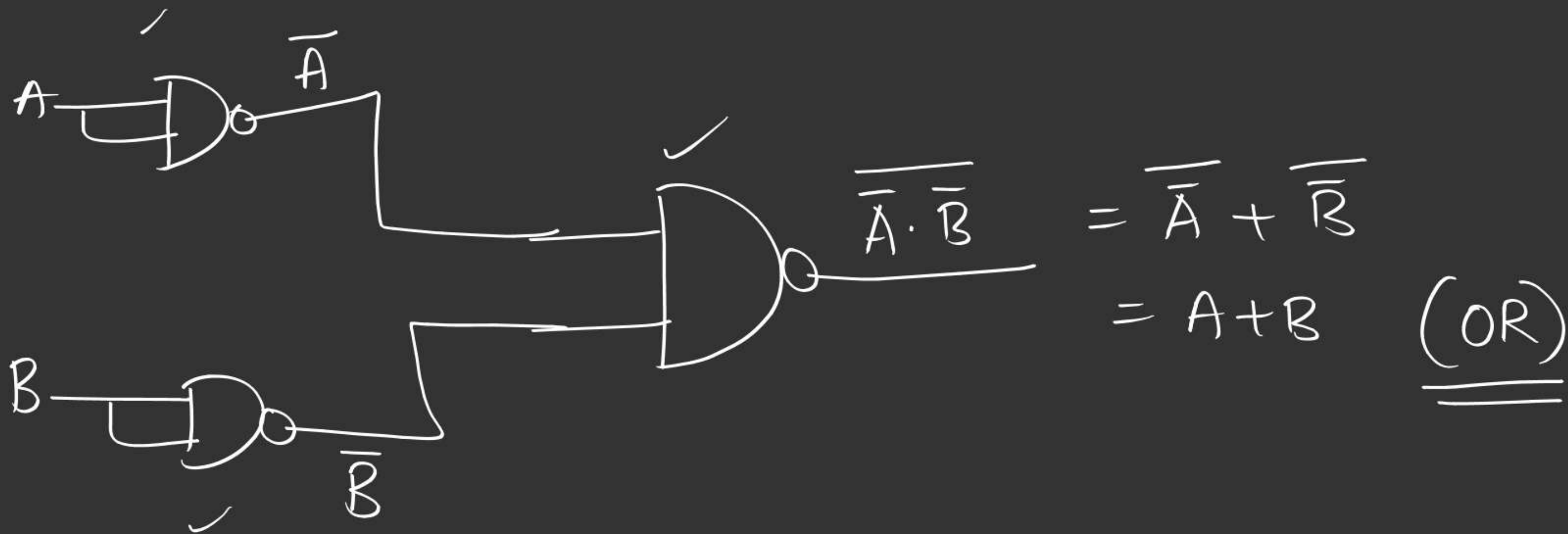
2) OR ✓



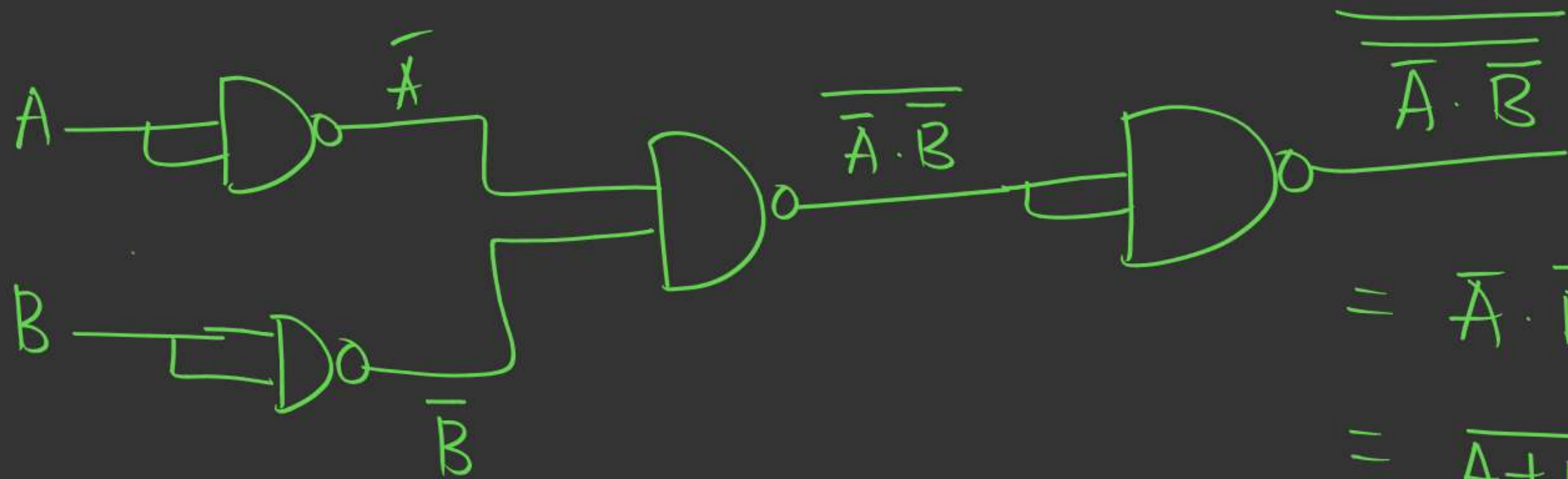
$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

$$\overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$$





(Bonus) NOR using NAND



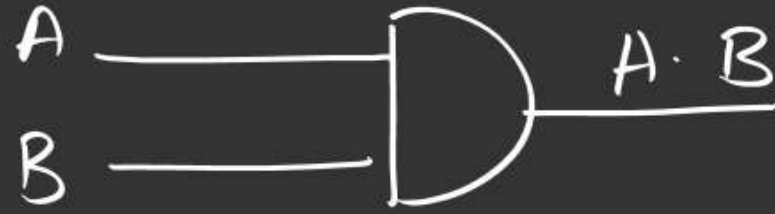
$$= \overline{\bar{A} \cdot \bar{B}}$$

$$= \overline{A+B} \Rightarrow \underline{\underline{\text{NOR}}}$$

3) AND



↳ (NAND + NOT)



$$y = A \oplus B = \bar{A}B + A\bar{B}$$

$$P = \bar{A}B \Rightarrow \bar{A} \rightarrow \bar{A} \cdot B \rightarrow \text{NOT}_{(1)} + \text{AND}_{(2)}$$

$$Q = A\bar{B} \Rightarrow \bar{B} \rightarrow A \cdot \bar{B} \rightarrow \text{NOT}_{(1)} + \text{AND}_{(2)}$$

$$y = P + Q \rightarrow \text{OR}_{(3)}$$



NOT $\rightarrow 1$

AND $\rightarrow 2$

OR $\rightarrow 3$

Total NAND = 9

Concept :-

$$y = PQ + RQ$$

$$\overline{y} = \overline{PQ + RQ}$$

$$\overline{y} = \overline{PQ} \cdot \overline{RQ}$$

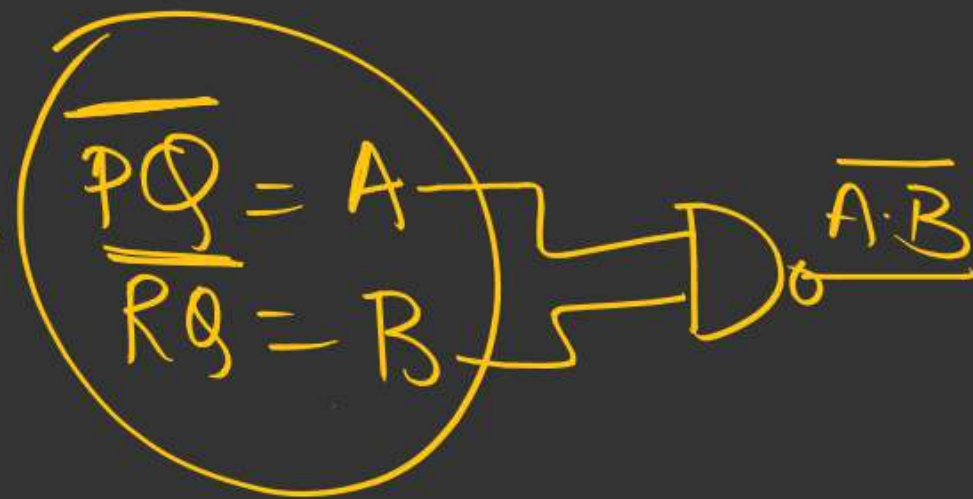
$$\overline{y} = \overline{PQ} \cdot \overline{RQ}$$

$$y = \overline{\overline{PQ} \cdot \overline{RQ}}$$

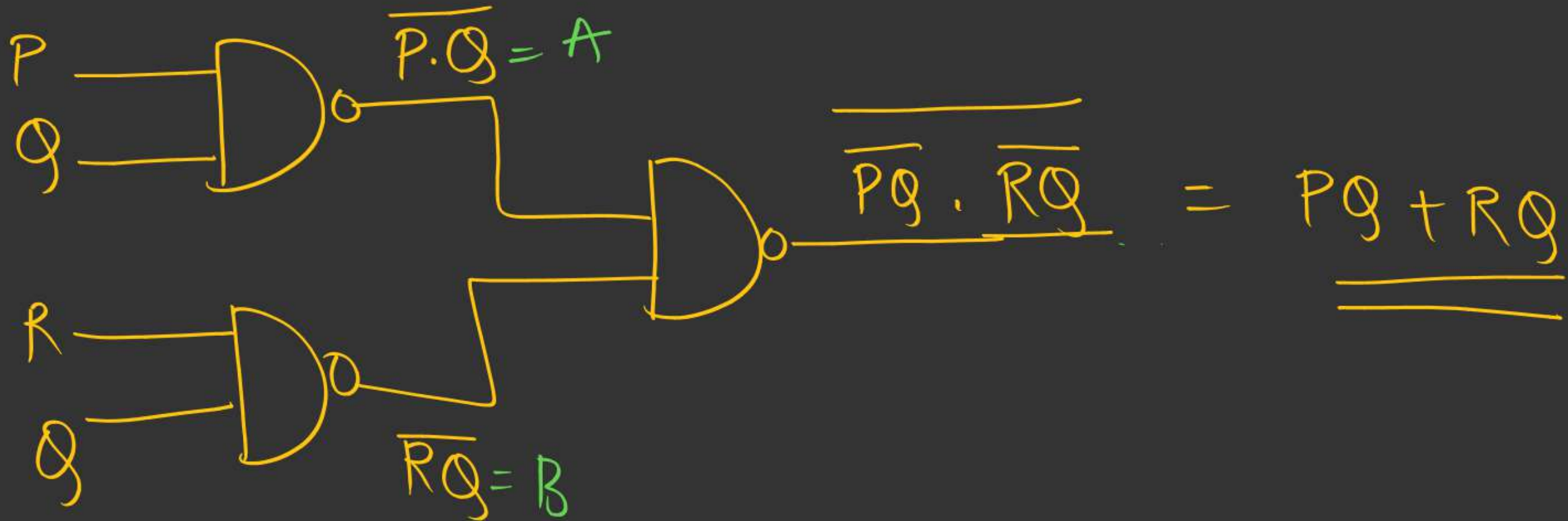
$$y = \overline{A \cdot B}$$

OR $\rightarrow$ 3
AND $\rightarrow$ 2
NOT $\rightarrow$ 1

3 NAND gates



$$y = \overline{PQ} \cdot \overline{RQ} = PQ + RQ$$



$$y = A \oplus B = \bar{A}B + A\bar{B}$$

$$\bar{y} = \overline{\bar{A}B + A\bar{B}}$$

$$\bar{y} = \overline{\bar{A}B} \cdot \overline{A\bar{B}}$$

$$\bar{y} = \overline{\bar{A} \cdot B} \cdot \overline{A \cdot \bar{B}}$$

$$y = \overline{\bar{A} \cdot B} \cdot \overline{A \cdot \bar{B}}$$

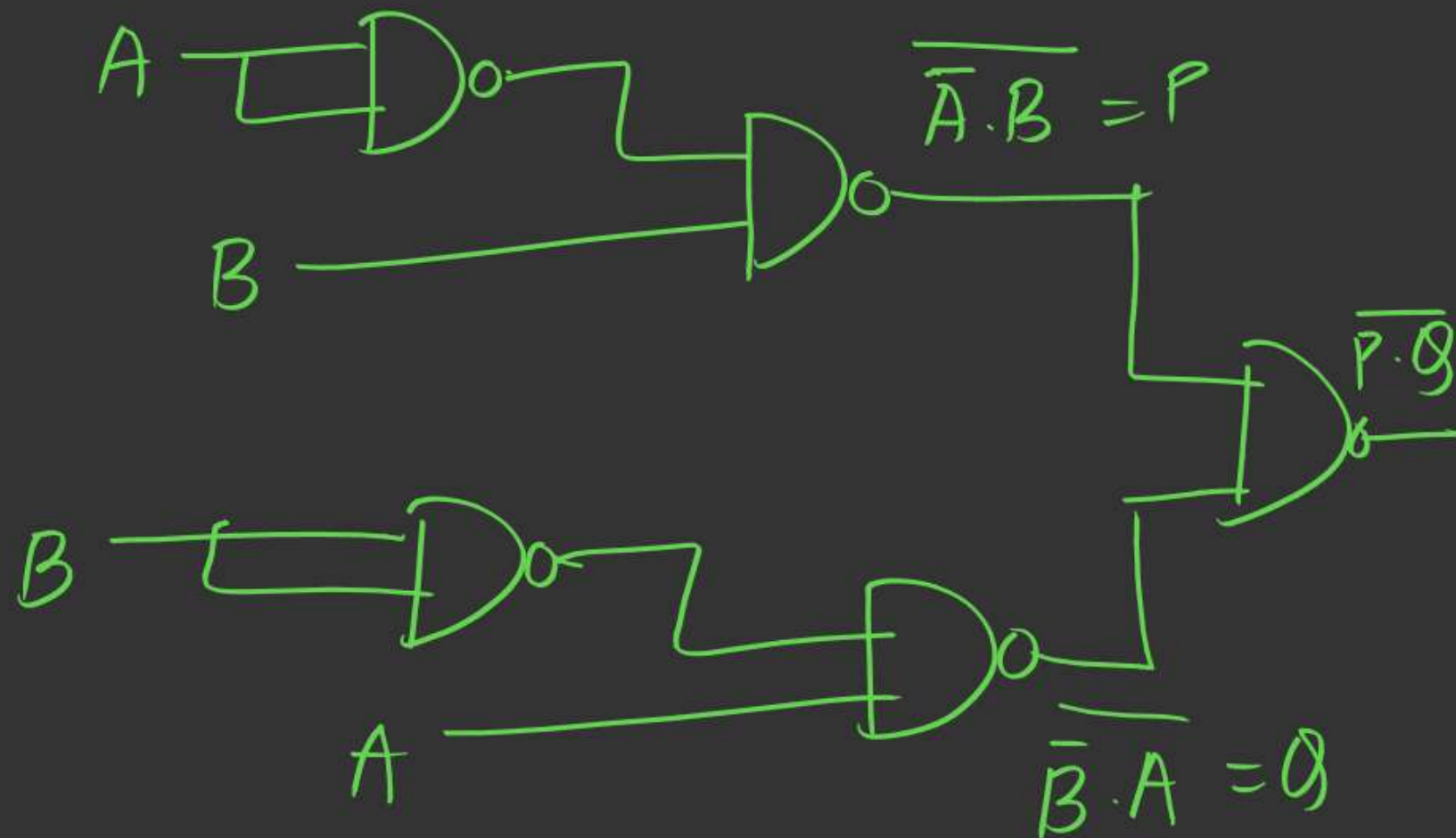
$$y = \overline{\overline{A} \cdot B} \cdot \overline{A \cdot \overline{B}} = A \oplus B$$

$$P = \overline{\overline{A} \cdot B} \rightarrow (2)$$

$$Q = \overline{A \cdot \overline{B}} \rightarrow (2)$$

$$y = \overline{P \cdot Q} \rightarrow (1)$$

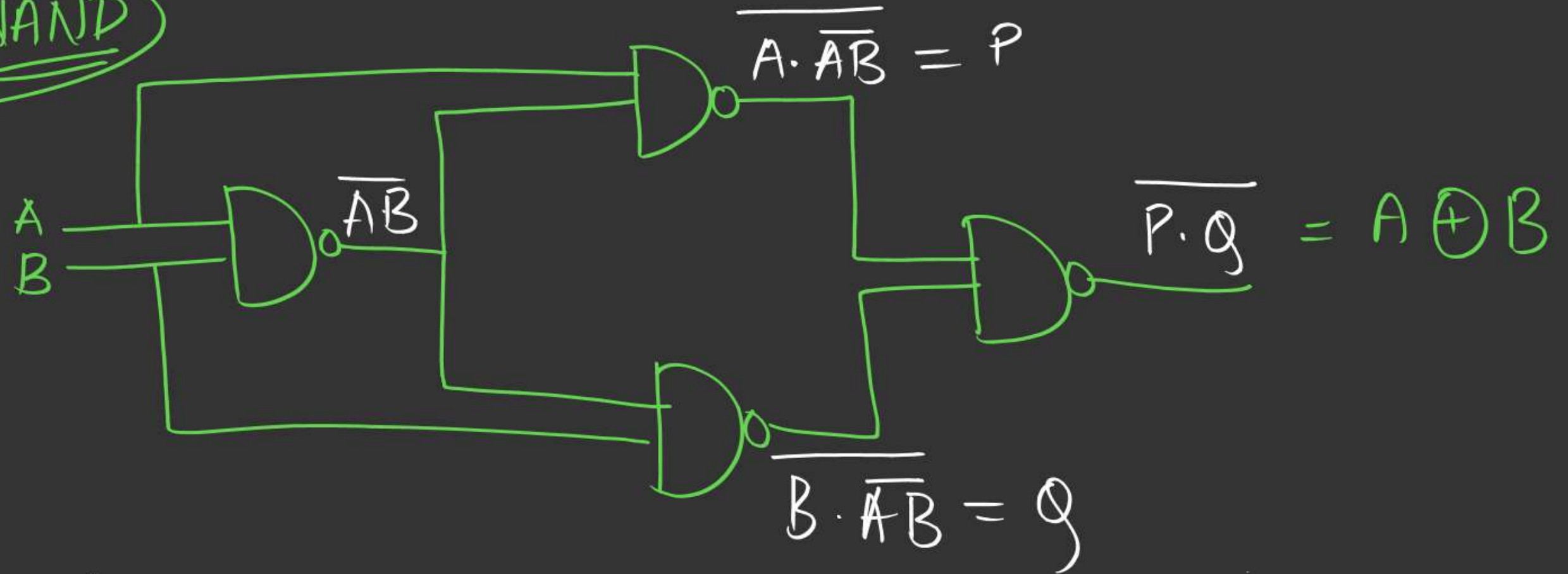
5 NAND



4) XOR



4 NAND



$$y = \overline{\underbrace{A \cdot \overline{AB}}_P} \cdot \overline{\underbrace{B \cdot \overline{AB}}_Q} = \underline{A \oplus B}$$

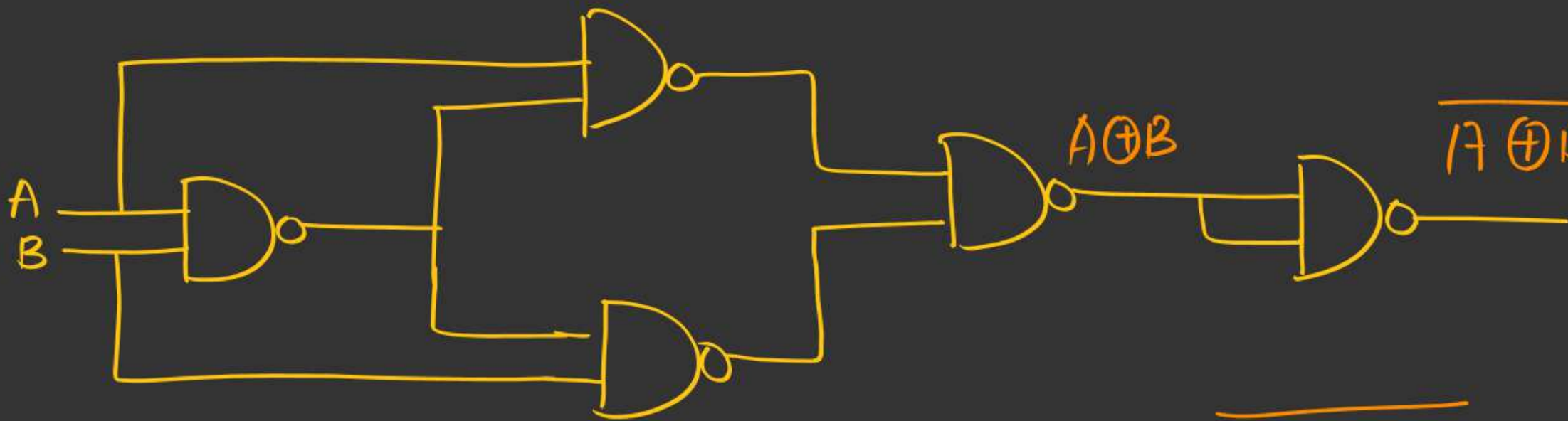
$\Rightarrow \overline{P \cdot Q}$

$$1) P = \overline{A \cdot \overline{AB}} = \overline{A} + \overline{\overline{AB}} = \overline{A} + AB = (\overline{A} + A) \cdot (\overline{A} + B) = 1 \cdot (\overline{A} + B) = (\overline{A} + B)$$

$$2) Q = \overline{B \cdot \overline{AB}} = \overline{B} + \overline{\overline{AB}} = \overline{B} + AB = (\overline{B} + A) (\overline{B} + B) = (\overline{B} + A)$$

$$3) y = \overline{P \cdot Q} = \overline{(\overline{A} + B)(\overline{B} + A)} = \overline{(\overline{A} + B)} + \overline{(\overline{B} + A)} = A \cdot \overline{B} + B \cdot \overline{A} = \overline{A}B + A\overline{B} = \underline{\underline{A \oplus B}}$$

5) XNOR $\rightarrow$ XOR + NOT
 (4) (1) = 5 NAND



$$\overline{A \oplus B} = A \odot B$$

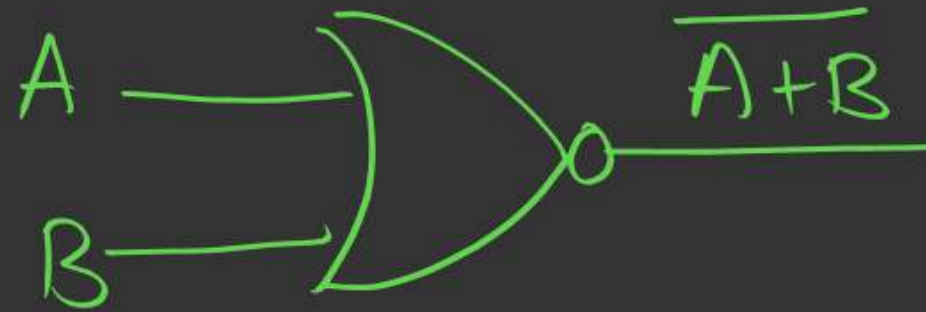
$$\overline{\overline{A}B + A\overline{B}}$$

$$= \overline{\overline{A}B} \cdot \overline{A\overline{B}} = (A + \overline{B})(\overline{A} + B)$$

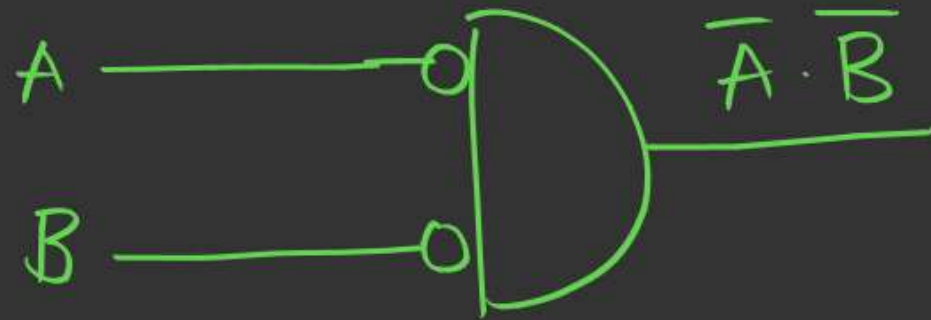
$$= \overline{\overline{A}\overline{B}} + \overline{AB} = \underline{A \odot B}$$

NOR Gate

Representation:



$$\overline{A+B} = \overline{A} \cdot \overline{B}$$



Truth Table:

	A	B	$\overline{A+B}$
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	0

$$\text{SOP} \Rightarrow \sum (0)$$

$$= \overline{A} \cdot \overline{B}$$

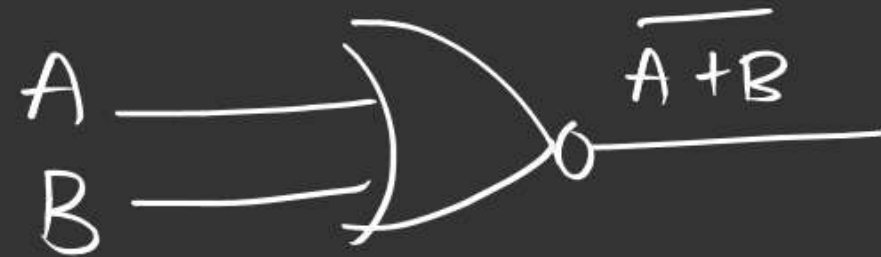
$$= \overline{A+B}$$

Obs:-

1) If any i/p is 1, NOR o/p is 0.

2) Only when all i/p's are 0, NOR o/p is 1.

Commutative Law:-

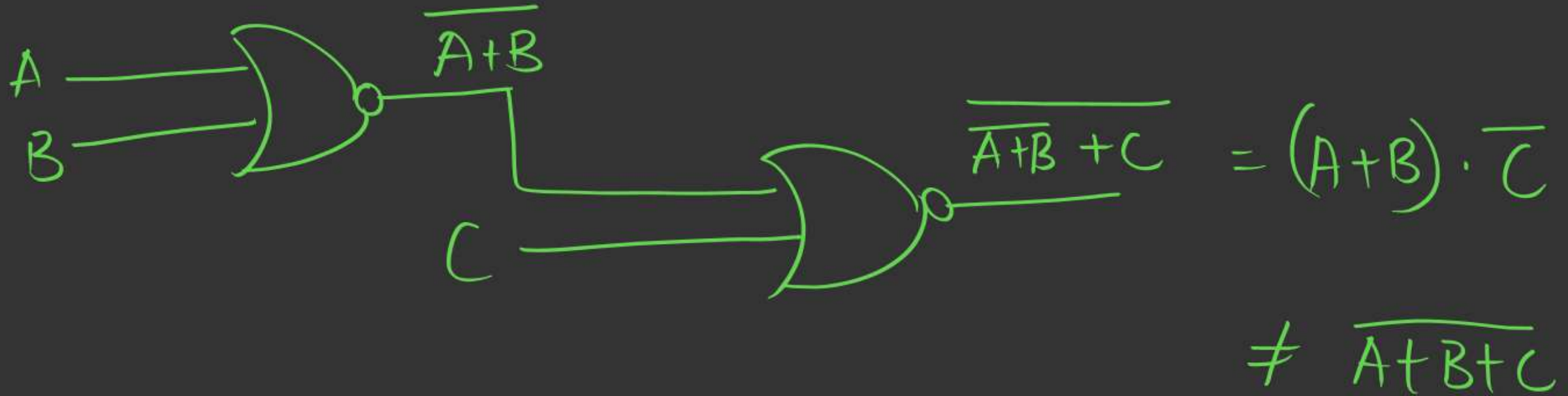
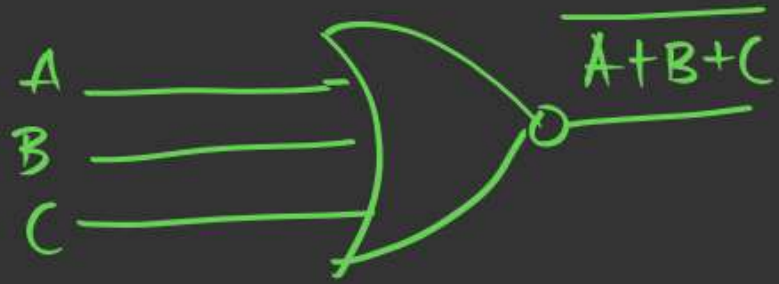


$$\overline{A+B} = \overline{B+A}$$



Holds.

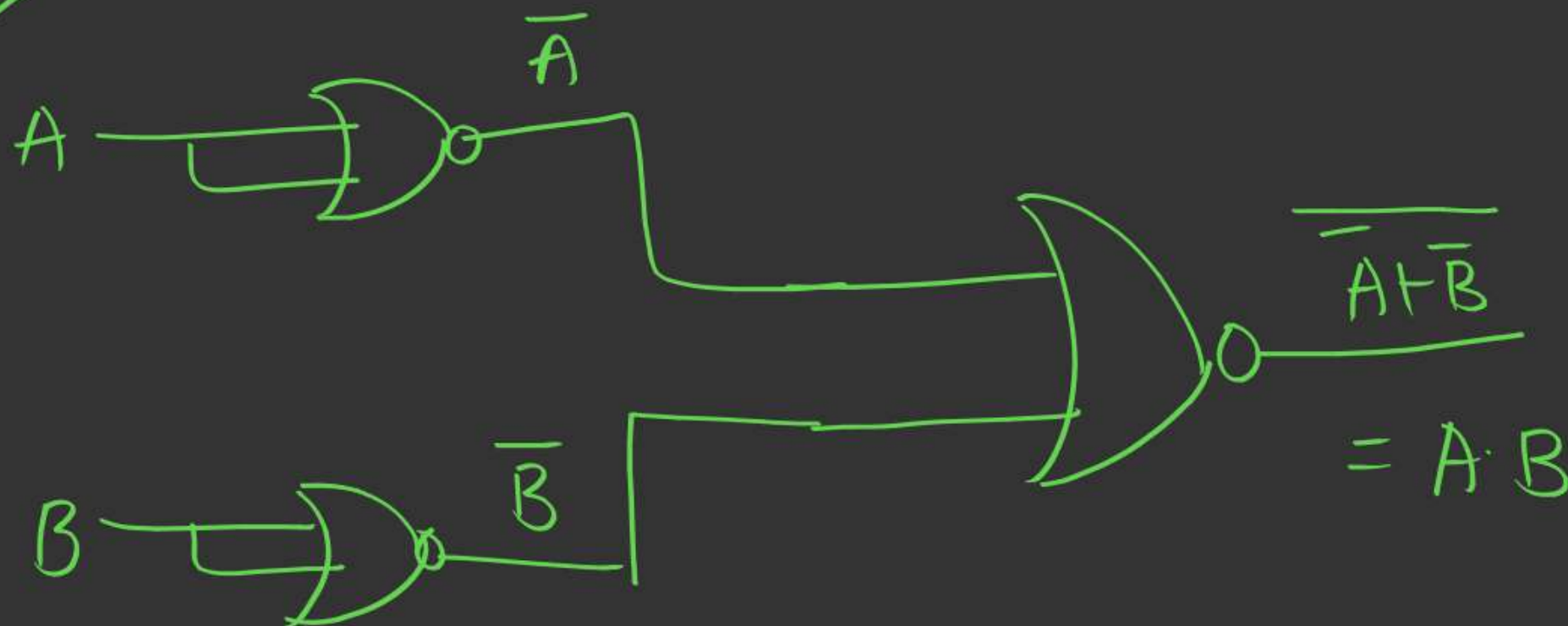
Associative law:-



Not Holds

3 NOR

2) AND gate ✓



$A \cdot B$

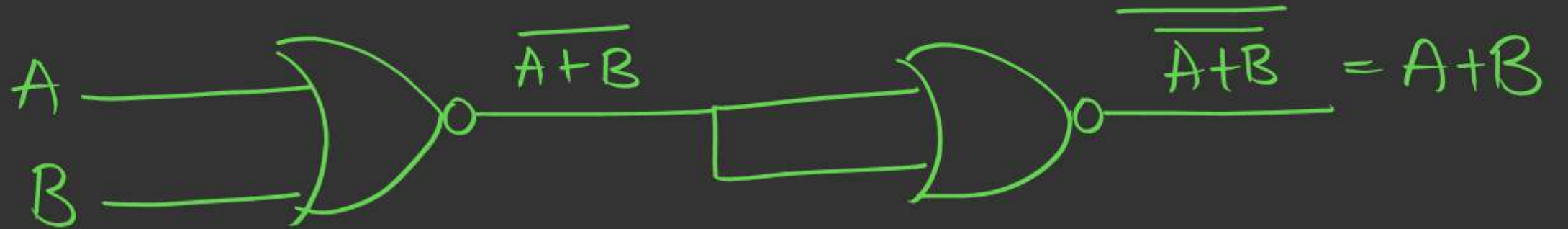
$$\overline{\overline{A} + \overline{B}} = A \cdot B$$

3) OR gate ✓

OR + NOT $\rightarrow$ NOR

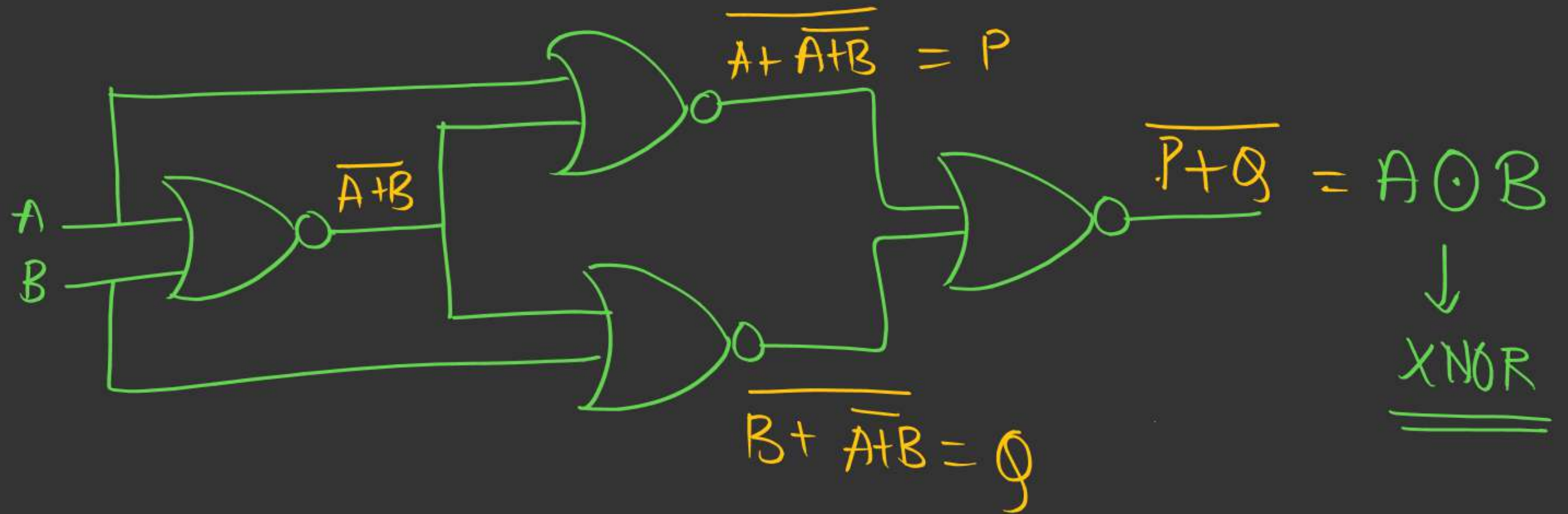
NOR + NOT $\rightarrow$ OR

2 NOR gates



4) XNOR using NOR

4 NOR gates



$$y = \overline{P+Q}$$

$$= \overline{A + \overline{A+B}} + \overline{B + \overline{A+B}}$$

$$1) P = \overline{A + \overline{A+B}} = \overline{A} \cdot (A+B) = \overline{A} \cdot A + \overline{A} \cdot B = \overline{A} \cdot B$$

$$2) Q = \overline{B + \overline{A+B}} = \overline{B} \cdot (A+B) = \overline{B} \cdot A + \overline{B} \cdot B = A \overline{B}$$

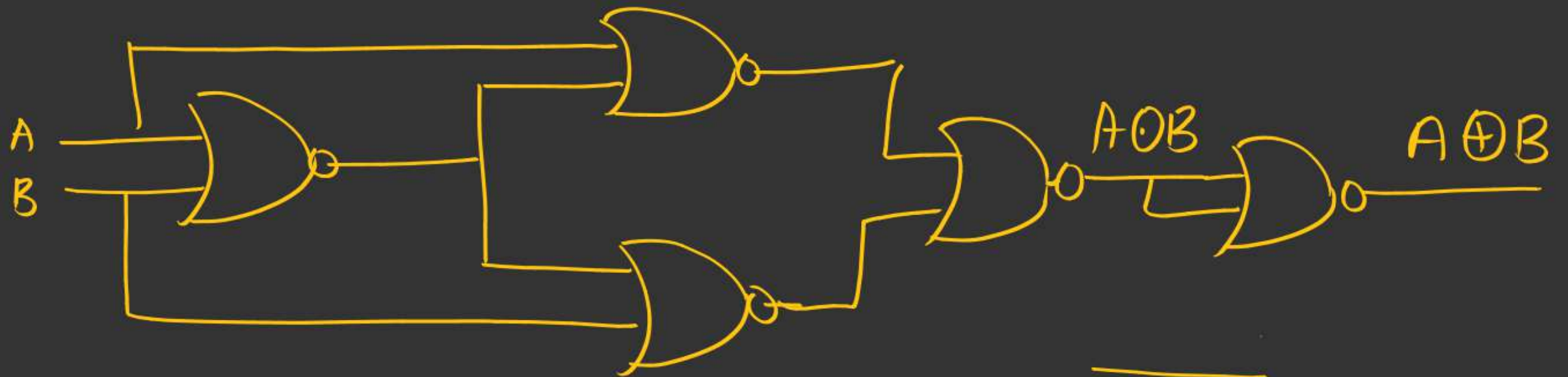
$$3) \overline{P+Q} = \overline{\overline{A}B + A\overline{B}} = \overline{A \oplus B} = A \odot B$$

5) XOR using NOR

$$XOR = XNOR + NOT$$

$$(4) \quad (1) = \underline{\underline{5 \text{ NOR}}}$$

5 NOR gates



$$\overline{\overline{A \cdot B}} = A \oplus B$$

Optimal/min no. of 2 i/p gates

★ Summary :

V.V. Imp

	2 i/p	2 i/p
GATES	No. of NAND	no. of NOR
NOT	1	1
AND	2	3
OR	3	2
XOR	4	5
XNOR	5	4



THANK - YOU